Metodi grafici per analisi dei meccanismi

Aronhold-Kennedy e diagrammi polari

Teorema di Aronhold Kennedy

"The Aronhold-Kennedy theorem: A graphical proof"
E. Pennestrì

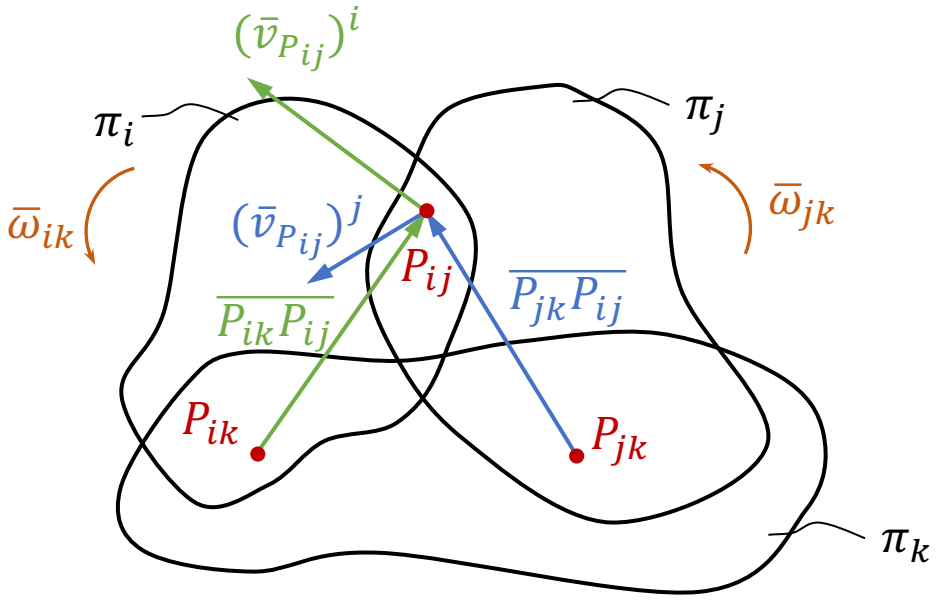
- Dimostrazione del teorema di Aronhold Kennedy
- Esercizi sul calcolo dei CIR e analisi cinematica tramite il teorema di Aronhold Kennedy

Teorema di Aronhold Kennedy

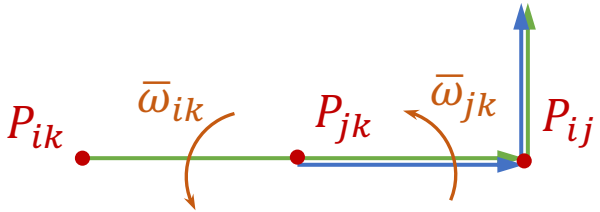
Dati tre corpi rigidi animati da moto piano, i CIR dei moti relativi devono essere allineati.

La dimostrazione viene condotta per assurdo, immaginando che uno dei tre CIR non sia allineato agli altri due.

Teorema di Aronhold Kennedy



$$(\bar{v}_{P_{ij}})^i = \bar{\omega}_{ik} \times \overline{P_{ik}P_{ij}}$$
$$(\bar{v}_{P_{ij}})^j = \bar{\omega}_{jk} \times \overline{P_{jk}P_{ij}}$$

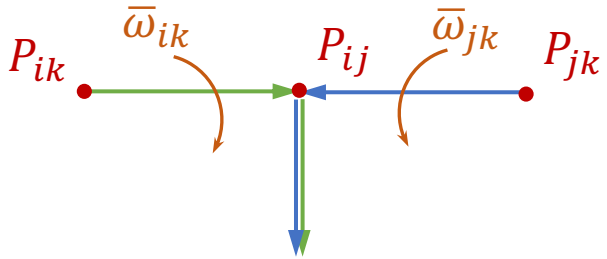


$$(\bar{v}_{P_{ij}})^i = (\bar{v}_{P_{ij}})^j$$

$$\bar{\omega}_{ik} \times \overline{P_{ik}P_{ij}} = \bar{\omega}_{jk} \times \overline{P_{jk}P_{ij}}$$

$$\omega_{ik}\, P_{ik}P_{ij} = \omega_{jk}\, P_{jk}P_{ij}$$

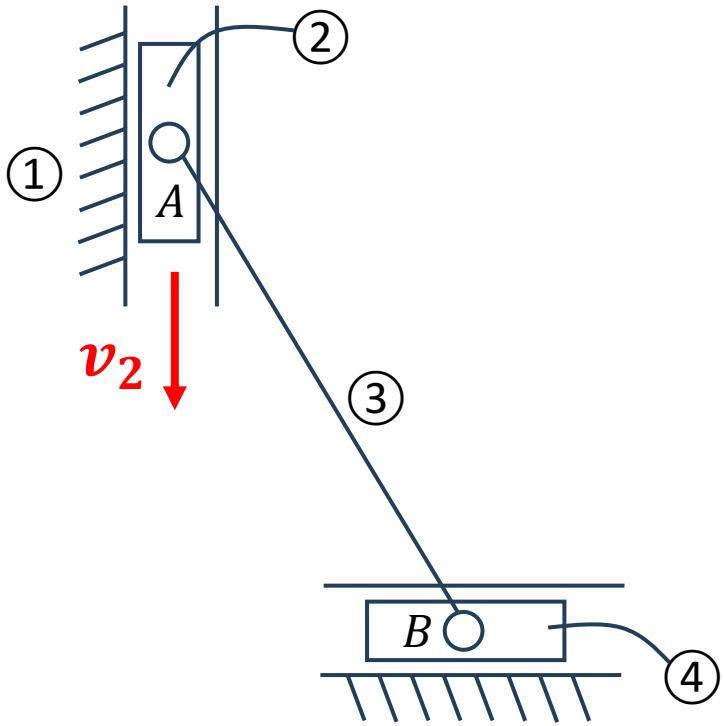
Se $\bar{\omega}_{ik}$ e $\bar{\omega}_{jk}$ fossero discordi



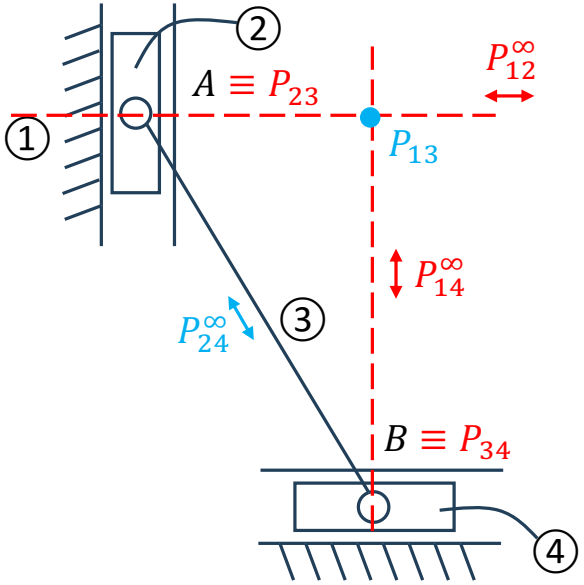
$$(\bar{v}_{P_{ij}})^i = (\bar{v}_{P_{ij}})^j$$

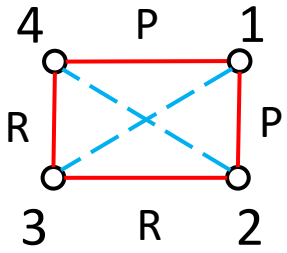
Analisi cinematica tramite teorema di Aronhold Kennedy

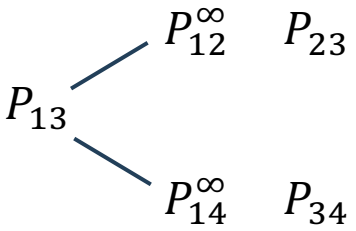
- Si determinino i CIR dei moti relativi, applicando il teorema di Aronhold-Kennedy.
- Ipotizzando di conoscere v_2, velocità del membro movente (2), si calcoli, nota completamente la geometria del meccanismo, la velocità del membro cedente (4).
- Si deduca, inoltre, quale delle due velocità è maggiore.

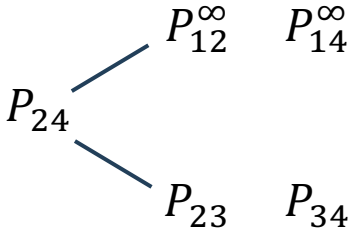


Analisi cinematica tramite teorema di Aronhold Kennedy









Applicando il teorema di Aronhold-Kennedy:

$$(\overrightarrow{v_{P_{23}}})^2 = (\overrightarrow{v_{P_{23}}})^3 \rightarrow \overrightarrow{v_2} = \overrightarrow{\omega_3} \times \overrightarrow{P_{13}P_{23}}$$

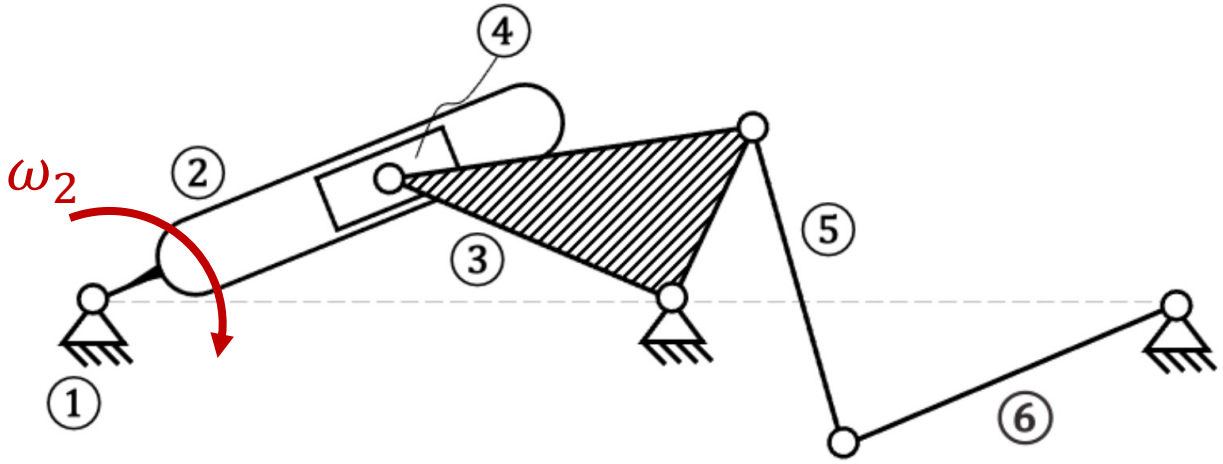
$$(\overrightarrow{v_{P_{34}}})^4 = (\overrightarrow{v_{P_{34}}})^3 \rightarrow \overrightarrow{v_4} = \overrightarrow{\omega_3} \times \overrightarrow{P_{13}P_{34}}$$

$$v_4 = v_2 \frac{\overline{P_{13}P_{34}}}{\overline{P_{13}P_{23}}}$$

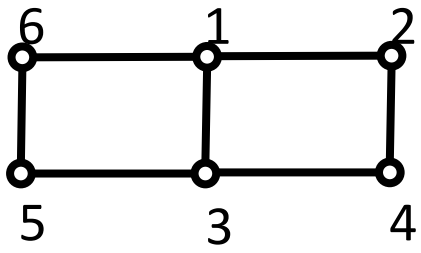
Poiché $\overline{P_{13}P_{34}} > \overline{P_{13}P_{23}}$ segue che $v_4 > v_2$

Analisi cinematica tramite teorema di Aronhold Kennedy

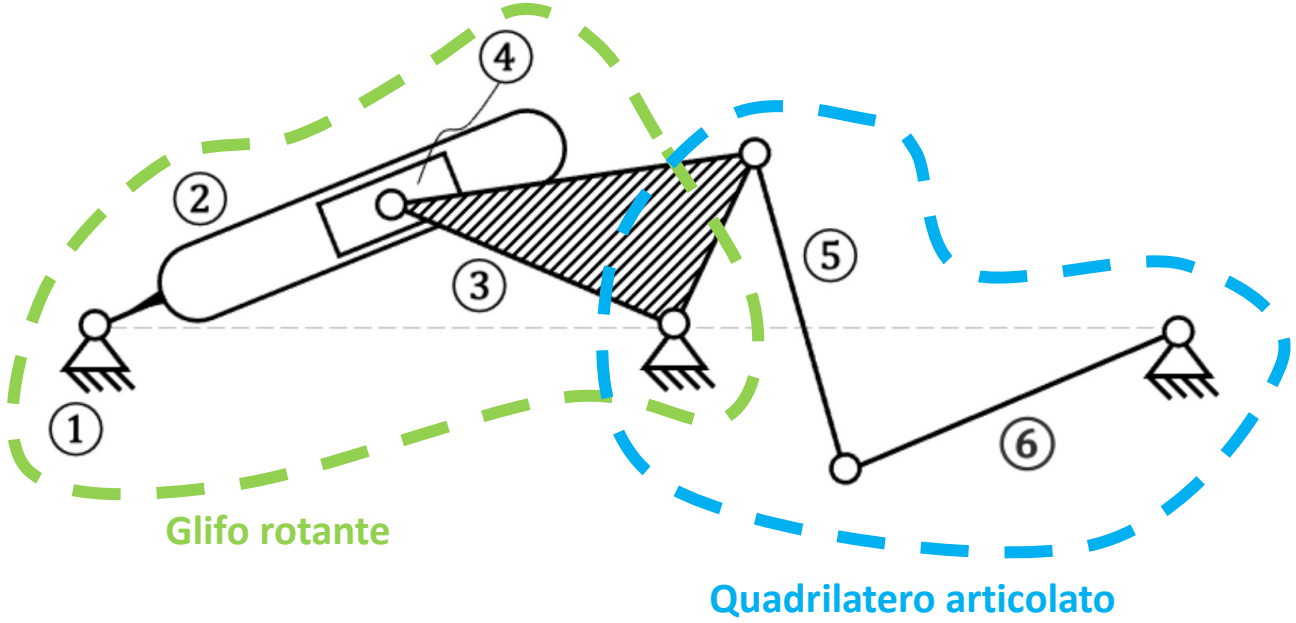
- Si determinino i CIR dei moti relativi, applicando il teorema di Aronhold-Kennedy.
- Ipotizzando di conoscere ω_2, velocità angolare del membro movente (2), si calcoli, nota completamente la geometria del meccanismo, ω_2, la velocità angolare del membro cedente (6).
- Si deduca, inoltre, quale delle due velocità è maggiore.



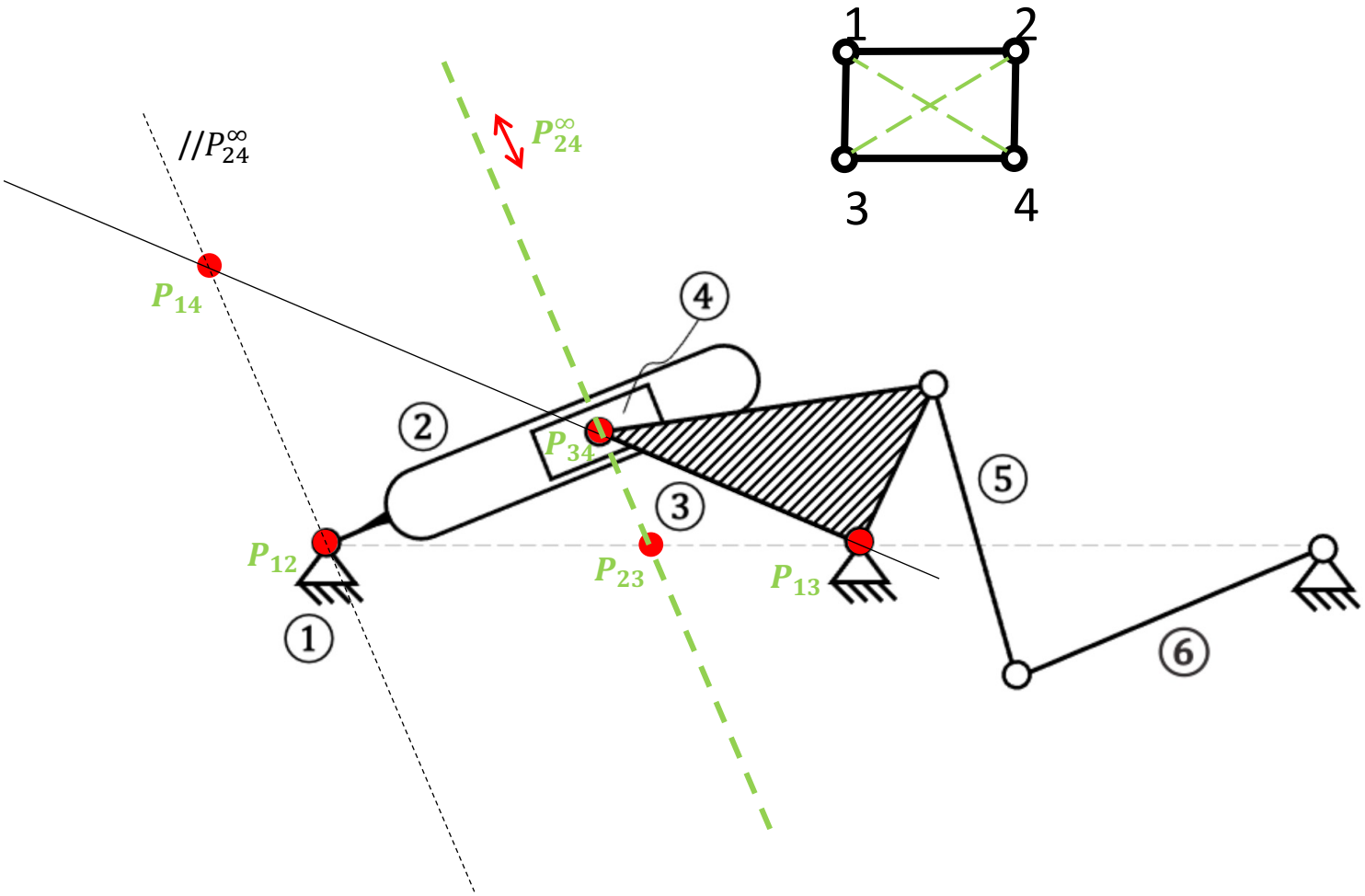
Analisi cinematica tramite teorema di Aronhold Kennedy



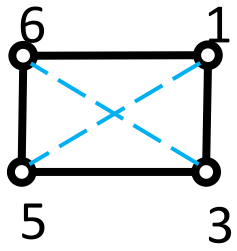
$$\#CIR = \frac{l(l-1)}{2} = 15$$

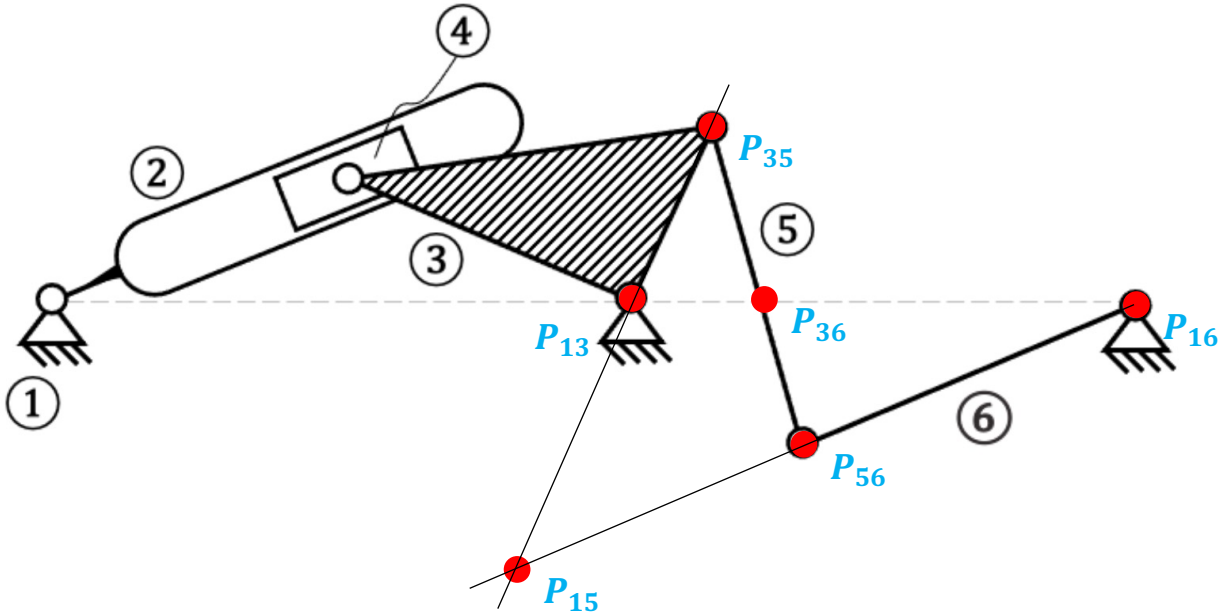


Analisi cinematica tramite teorema di Aronhold Kennedy

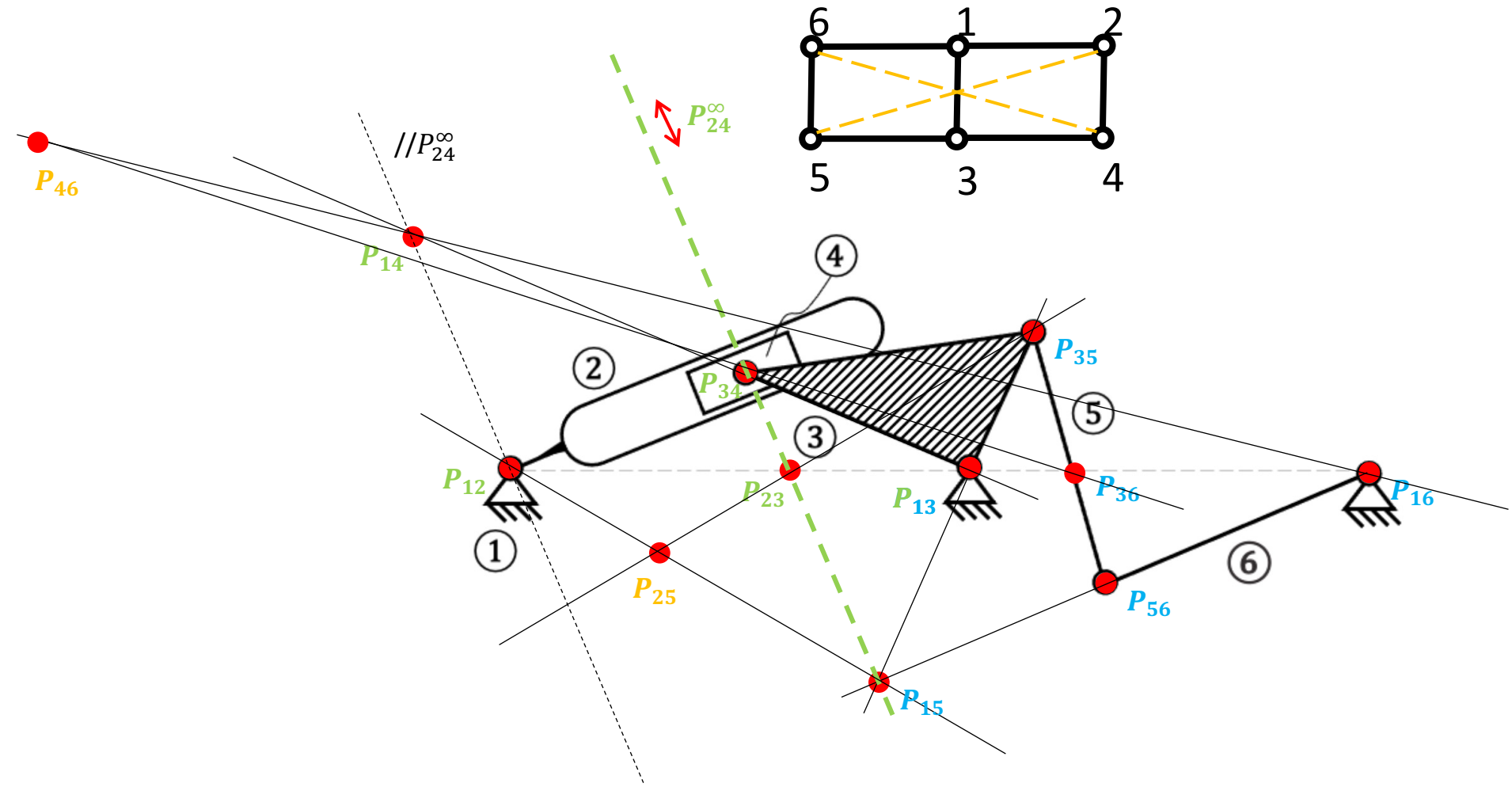


Analisi cinematica tramite teorema di Aronhold Kennedy

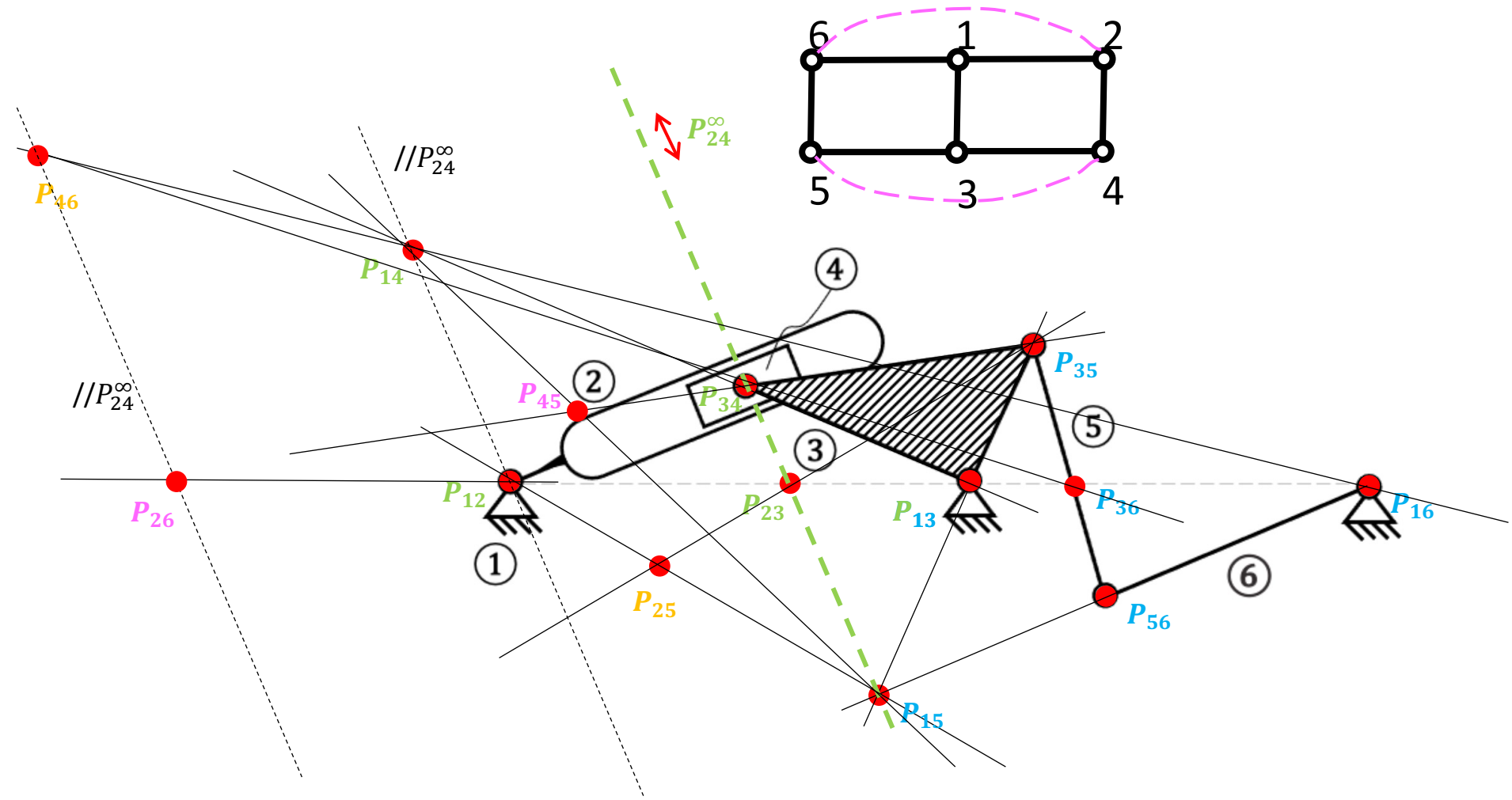




Analisi cinematica tramite teorema di Aronhold Kennedy



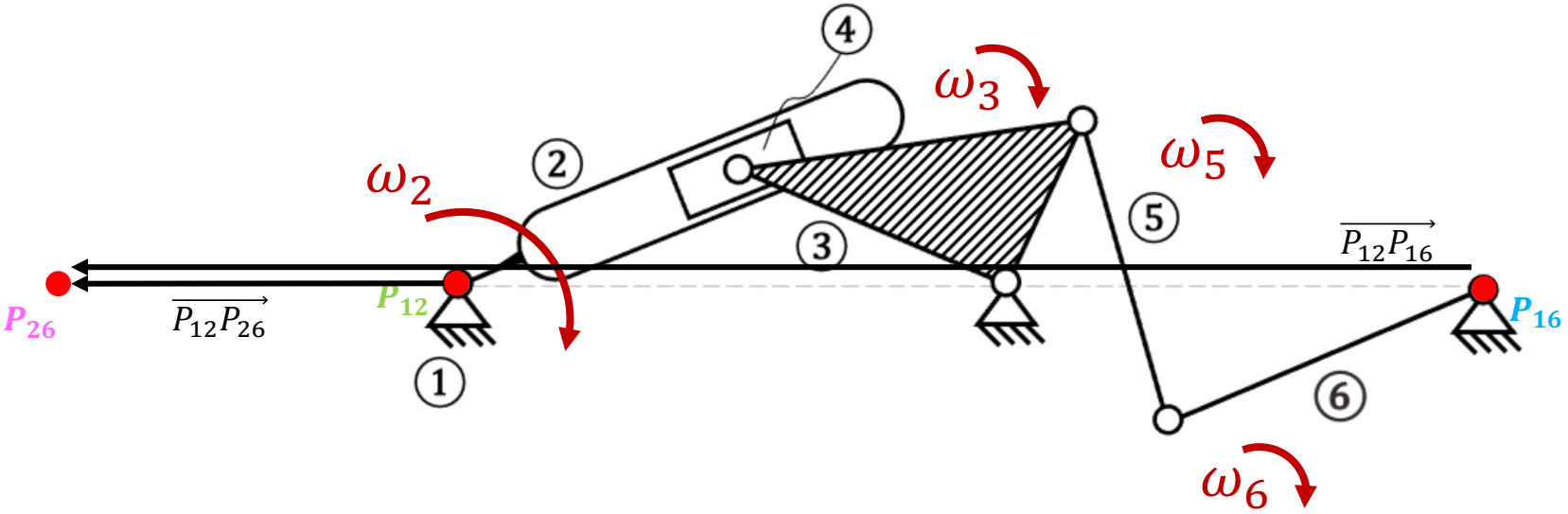
Analisi cinematica tramite teorema di Aronhold Kennedy



Analisi cinematica tramite teorema di Aronhold Kennedy

$$(\overrightarrow{v_{P_{26}}})^2 = (\overrightarrow{v_{P_{26}}})^6$$

$$\overrightarrow{\omega_2} \times \overrightarrow{P_{12}P_{26}} = \overrightarrow{\omega_6} \times \overrightarrow{P_{16}P_{26}} \qquad \rightarrow \omega_6 = \omega_2 \frac{\overline{P_{12}P_{26}}}{\overline{P_{16}P_{26}}}$$



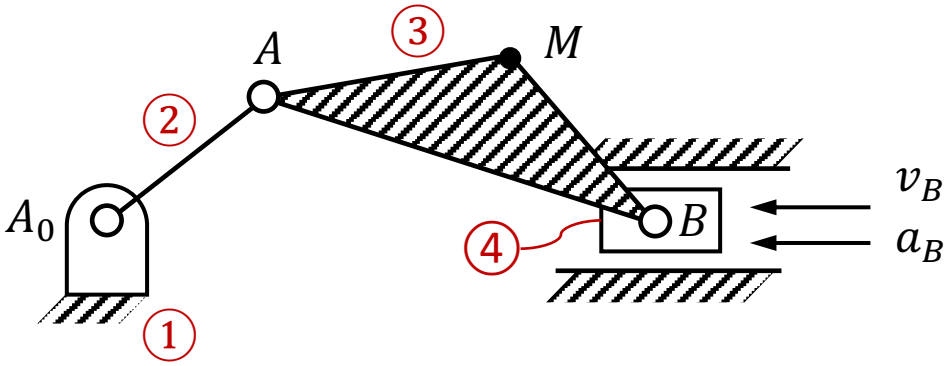
Poiché $P_{12}P_{26} < P_{16}P_{26}$ segue che $\omega_6 < \omega_2$

Analisi cinematica tramite metodo dei diagrammi polari

Introduzione alla Cinematica dei Meccanismi vol III
(Capito 8, tavole 1-21)

- Di seguito alcuni esempi

MDP: manovellismo di spinta



$$\bar{v}_A = \bar{v}_B + \bar{v}_{AB}$$

$$\bar{v}_A = \bar{\omega}_2 \times \overline{A_0A}$$

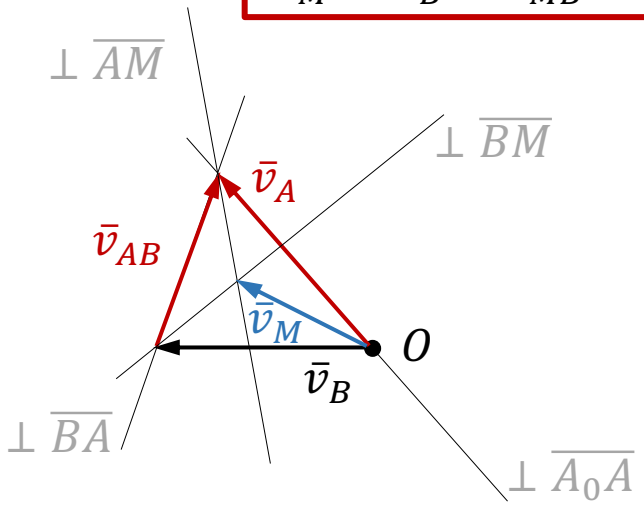
$$\bar{v}_{AB} = \bar{\omega}_3 \times \overline{BA}$$

Quantità note: $\bar{v}_B$, $\bar{a}_B$
Incognite: $\bar{v}_M$, $\bar{a}_M$

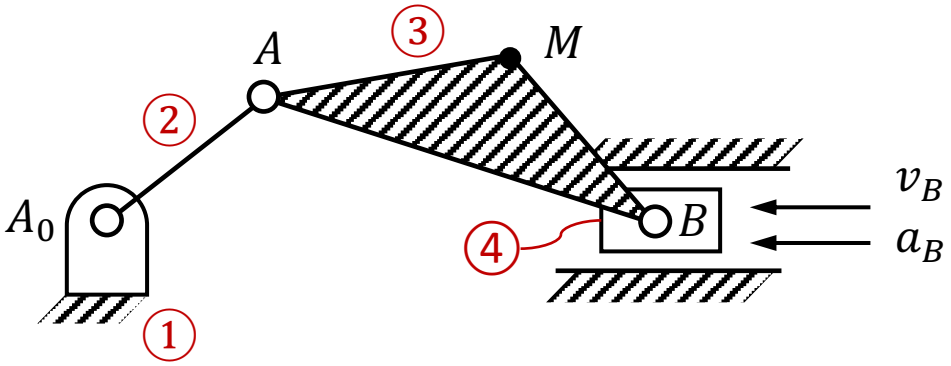
	MOD	DIR
$\bar{v}_A$	?	$\perp \overline{A_0A}$
$\bar{v}_B$	v_B	// patt
$\bar{v}_{AB}$	?	$\perp \overline{BA}$

$$\bar{v}_M = \bar{v}_A + \bar{v}_{MA} = \bar{v}_A + \bar{\omega}_3 \times \overline{AM}$$

$$\bar{v}_M = \bar{v}_B + \bar{v}_{MB} = \bar{v}_B + \bar{\omega}_3 \times \overline{BM}$$



MDP: manovellismo di spinta



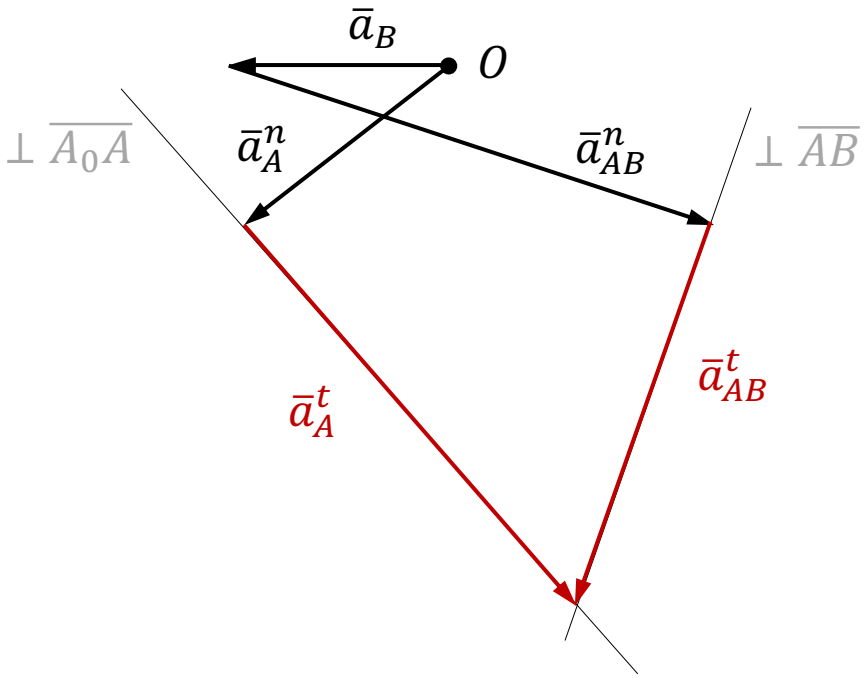
$$\bar{a}_A = \bar{a}_B + \bar{a}_{AB}$$

$$\bar{a}_A^n + \bar{a}_A^t = \bar{a}_B + \bar{a}_{AB}^n + \bar{a}_{AB}^t$$

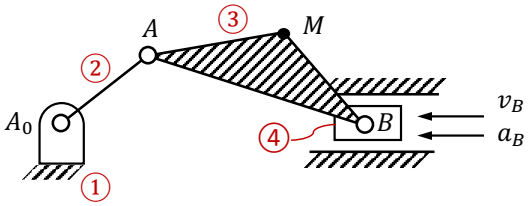
Quantità note: $\bar{v}_B$, $\bar{a}_B$

Incognite: $\bar{v}_M$, $\bar{a}_M$

	MOD	DIR
$\bar{a}_A^n$	$\omega_2^2 AA_0$	// $\overline{AA_0}$
$\bar{a}_A^t$	?	⊥ $\overline{A_0A}$
$\bar{a}_B$	a_B	// patt
$\bar{a}_{AB}^n$	$\omega_3^2 AB$	// $\overline{AB}$
$\bar{a}_{AB}^t$	?	⊥ $\overline{AB}$

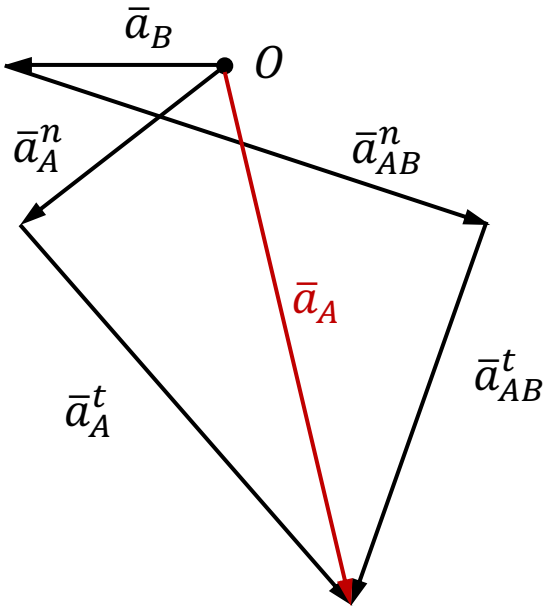


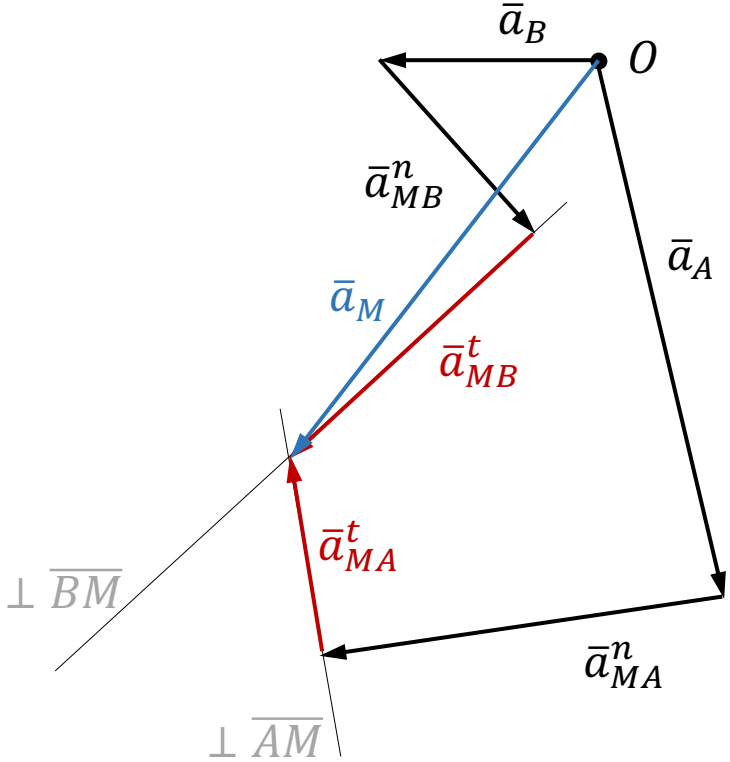
MDP: manovellismo di spinta



$$\bar{a}_M = \bar{a}_A + \bar{a}_{MA} = \bar{a}_A + \bar{a}^n_{MA} + \bar{a}^t_{MA} = \bar{a}_A + \omega_3^2 \overline{MA} + \bar{\alpha}_3 \times \overline{AM}$$

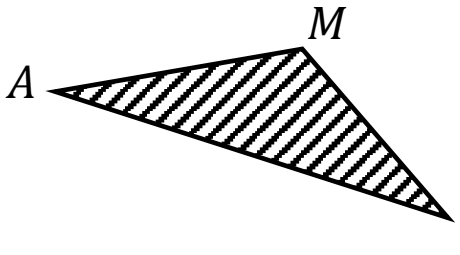
$$\bar{a}_M = \bar{a}_B + \bar{a}_{MB} = \bar{a}_B + \bar{a}^n_{MB} + \bar{a}^t_{MB} = \bar{a}_B + \omega_3^2 \overline{MB} + \bar{\alpha}_3 \times \overline{BM}$$



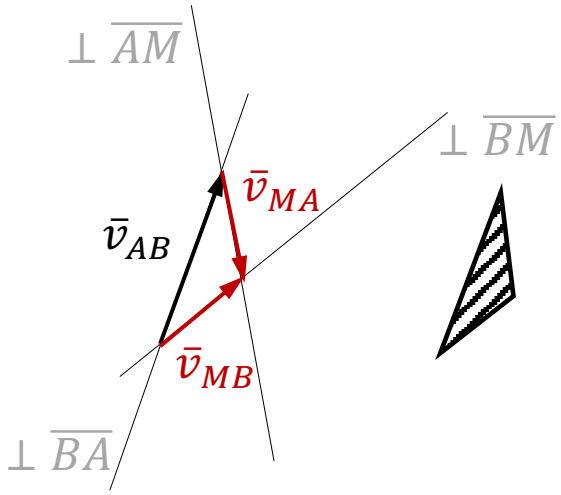


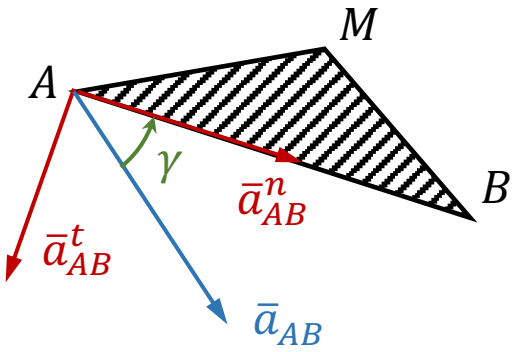
MDP: manovellismo di spinta

Opzioni alternative

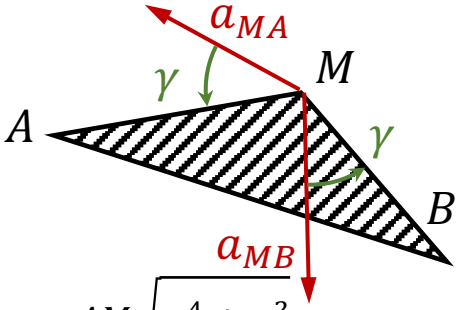


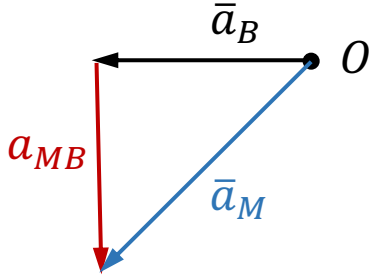
$$\bar{v}_{AB} = \bar{\omega}_3 \times \overline{BA} \rightarrow v_{AB} \propto BA$$
$$\bar{v}_{MA} = \bar{\omega}_3 \times \overline{AM} \rightarrow v_{MA} \propto AM$$
$$\bar{v}_{MB} = \bar{\omega}_3 \times \overline{BM} \rightarrow v_{MB} \propto BM$$





$$\tan\gamma = \frac{a^t_{AB}}{a^n_{AB}} = \frac{\alpha_3 AB}{\omega_3^2 AB} = \frac{\alpha_3}{\omega_3^2}$$ Invariante!

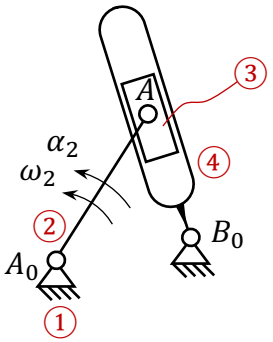




$$a_{MA} = AM\sqrt{\omega_3^4 + \alpha_3^2}$$
$$a_{MB} = MB\sqrt{\omega_3^4 + \alpha_3^2}$$

MDP: glifo rotante

Se $A_0A \geq A_0B_0 \rightarrow$ glifo **rotante**, altrimenti **oscillante**



$$\bar{\omega}_3 = \bar{\omega}_4 + \bar{\omega}_{34}$$

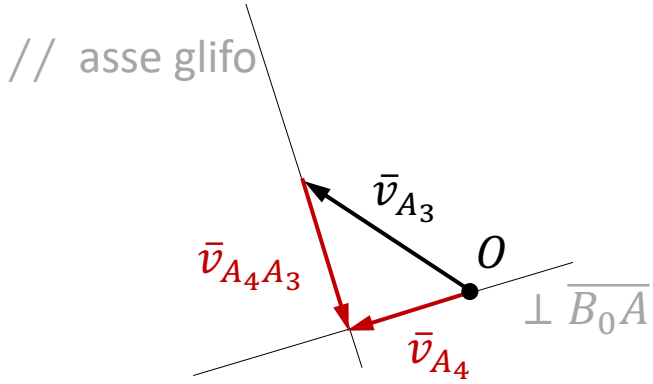
$$\bar{v}_{A_2} = \bar{v}_{A_3} = \bar{\omega}_2 \times \overline{A_0A}$$

$$\bar{v}_{A_4} = \bar{v}_{A_3} + \bar{v}_{A_4A_3}$$

$$\bar{v}_{A_4} = \bar{\omega}_4 \times \overline{B_0A}$$

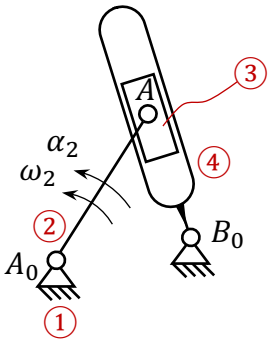
Quantità note: $\bar{\omega}_2$, $\bar{\alpha}_2$
Incognite: $\bar{\omega}_4$, $\bar{\alpha}_4$

	MOD	DIR
$\bar{v}_{A_4}$	?	$\perp \overline{B_0A}$
$\bar{v}_{A_3}$	$\omega_2 A_0A$	$\perp \overline{A_0A}$
$\bar{v}_{A_4A_3}$	?	// glifo



$$\omega_4 = \frac{v_{A_4}}{B_0A}$$

MDP: glifo rotante



$$\bar{a}_{A_4} = \bar{a}_{A_3} + \bar{a}_{A_4A_3} + \bar{a}_{cor}$$

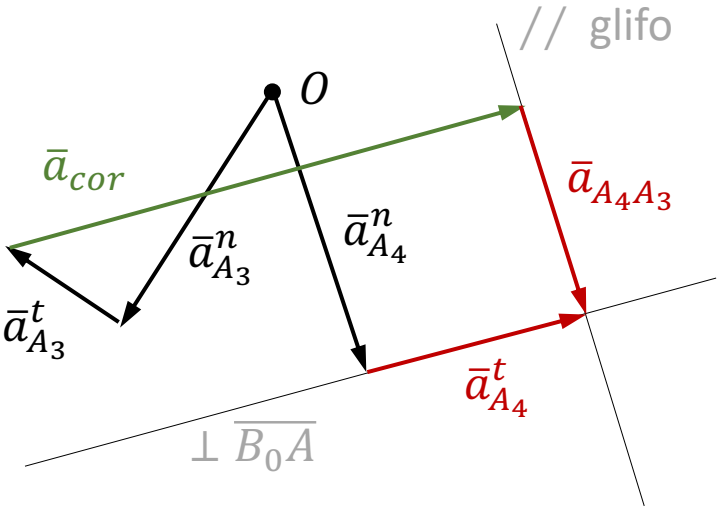
$$\bar{\omega}_3 = \bar{\omega}_4$$

$$\bar{a}_{cor} = 2\,\bar{\omega}_4 \times \bar{v}_{A_4A_3} = 2\,\bar{\omega}_3 \times \bar{v}_{A_4A_3}$$

$$\bar{a}^n_{A_4} + \bar{a}^t_{A_4} = \bar{a}^n_{A_3} + \bar{a}^t_{A_3} + \bar{a}_{A_4A_3} + 2\,\bar{\omega}_3 \times \bar{v}_{A_4A_3}$$

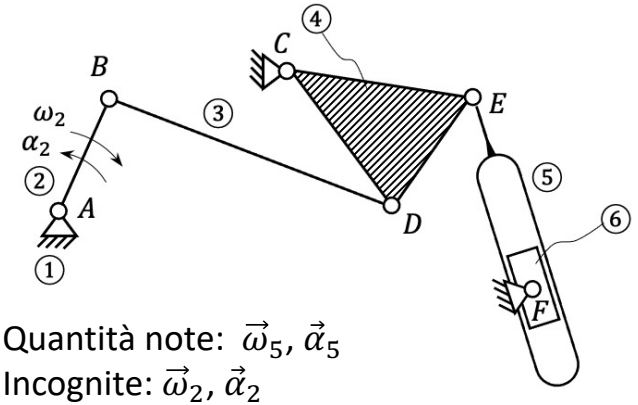
$\bar{\alpha}_4 \times \overline{B_0A}$ $\qquad$ $\bar{\alpha}_2 \times \overline{A_0A}$

	MOD	DIR
$\bar{a}^n_{A_4}$	$\omega_4^2 AB_0$	// $\overline{AB_0}$
$\bar{a}^t_{A_4}$	?	$\perp \overline{B_0A}$
$\bar{a}^n_{A_3}$	$\omega_2^2 AA_0$	// $\overline{AA_0}$
$\bar{a}^t_{A_3}$	$\alpha_2 A_0 A$	$\perp \overline{AA_0}$
$\bar{a}_{A_4A_3}$	?	// glifo
$\bar{a}_{cor}$	$2\omega_4 v_{A_4A_3}$	$\perp$ glifo



$$\alpha_4 = \alpha_3 = \frac{a^t_{A_4}}{B_0A}$$

MDP: glifo + quadrilatero



	MOD	DIR
v_D	?	$\perp \overline{CD}$
v_B	$\omega_2 AB$	$\perp \overline{AB}$
v_{DB}	?	$\perp \overline{BD}$

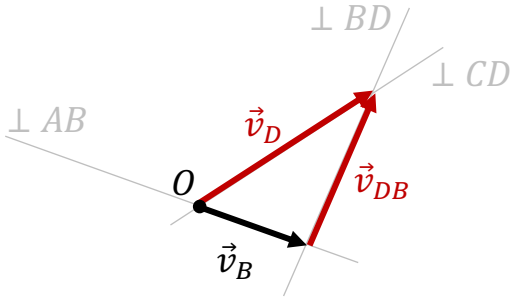
	MOD	DIR
v_F	?	$//\overline{EF}$
v_E	$\omega_4 CE$	$\perp \overline{CE}$
v_{FE}	?	$\perp \overline{EF}$

Quantità note: $\vec{\omega}_5, \vec{\alpha}_5$
Incognite: $\vec{\omega}_2, \vec{\alpha}_2$

$\vec{v}_D = \vec{v}_B + \vec{v}_{DB} \rightarrow \vec{\omega}_4 \times \overrightarrow{CD} = \vec{\omega}_2 \times \overrightarrow{AB} + \vec{\omega}_3 \times \overrightarrow{BD}$

Essendo $\vec{v}_D = \vec{\omega}_4 \times \overrightarrow{CD} \rightarrow \omega_4 = \dfrac{|\vec{v}_D|}{\overline{CD}}$ ↻

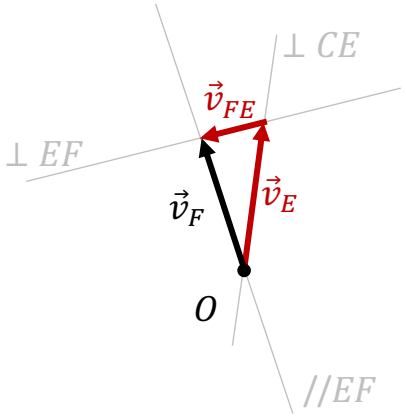
Essendo $\vec{v}_{DB} = \vec{\omega}_3 \times \overrightarrow{BD} \rightarrow \omega_3 = \dfrac{|\vec{v}_{DB}|}{\overline{BD}}$ ↻

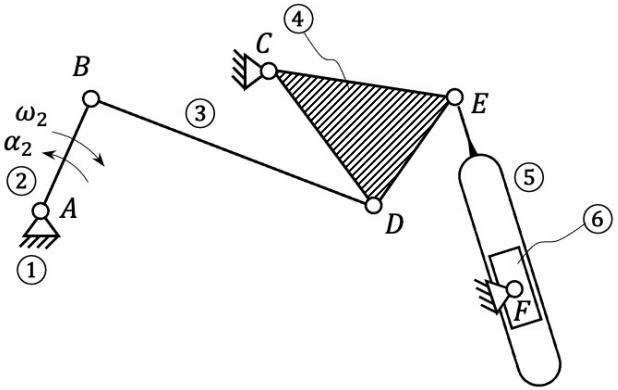


$\vec{v}_F = \vec{v}_E + \vec{v}_{FE} \rightarrow \vec{v}_F = \vec{\omega}_4 \times \overrightarrow{CE} + \vec{\omega}_5 \times \overrightarrow{EF}$

Essendo $\vec{v}_{FE} = \vec{\omega}_5 \times \overrightarrow{EF} \rightarrow \omega_5 = \dfrac{|\vec{v}_{FE}|}{\overline{EF}}$ ↻

Essendo $\vec{v}_{F_5} = \vec{v}_{F_6} + \vec{v}_{F_5F_6} \rightarrow \omega_6 = \omega_5$ ↻





	MOD	DIR
a_D^n	$\omega_4^2\ CD$	$//\overline{CD}$
a_D^t	?	$\perp DC$
a_B^n	$\omega_2^2\ BA$	$//\overline{BA}$
a_B^t	$\alpha_2\ AB$	$\perp \overline{AB}$
a_{DB}^n	$\omega_3^2\ DB$	$//\overline{DB}$
a_{DB}^t	?	$\perp \overline{DB}$

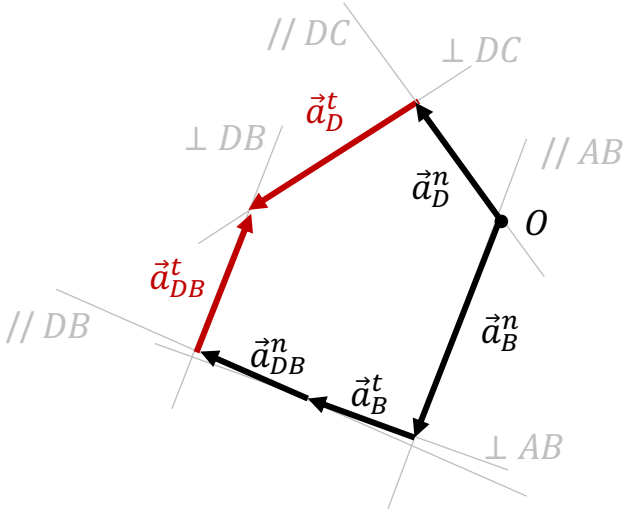
	MOD	DIR
a_E^n	$\omega_4^2\ EC$	$//\overline{EC}$
a_E^t	$\alpha_4\ CE$	$\perp \overline{CE}$
a_{FE}^n	$\omega_5^2\ FE$	$//\overline{FE}$
a_{FE}^t	?	$\perp \overline{FE}$
$a_{F_5F_6}$	?	$//\overline{FE}$
a_{cor}	$2\omega_4 v_F$	$\perp \overline{FE}$

$$\vec{a}_D = \vec{a}_B + \vec{a}_{DB} \rightarrow \vec{a}_D^n + \vec{a}_D^t = \vec{a}_B^n + \vec{a}_B^t + \vec{a}_{DB}^n + \vec{a}_{DB}^t$$

$$\rightarrow \omega_4^2\,\overrightarrow{CD} + \vec{\alpha}_4 \times \overrightarrow{CD} = \omega_2^2\,\overrightarrow{BA} + \vec{\alpha}_2 \times \overrightarrow{AB} + \omega_3^2\,\overrightarrow{DB} + \vec{\alpha}_3 \times \overrightarrow{BD}$$

Essendo $\vec{a}_D^t = \vec{\alpha}_4 \times \overrightarrow{CD}$ $\rightarrow$ $\alpha_4 = \dfrac{|\vec{a}_D^t|}{\overline{CD}}$

Essendo $\vec{a}_{DB}^t = \vec{\alpha}_3 \times \overrightarrow{BD}$ $\rightarrow$ $\alpha_3 = \dfrac{|\vec{a}_{DB}^t|}{\overline{BD}}$



$$\vec{a}_F = \vec{a}_E + \vec{a}_{FE}$$

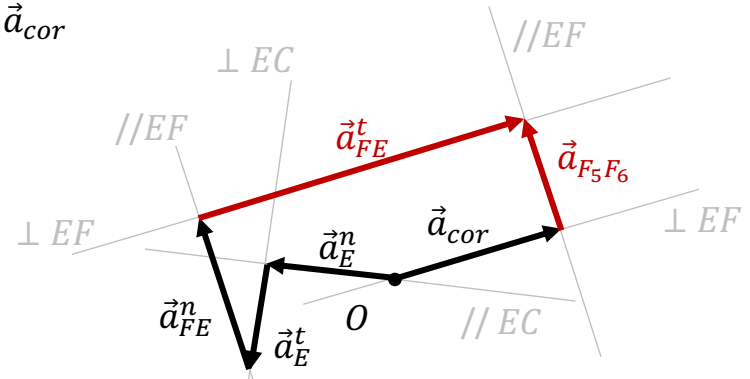
$$\vec{a}_{F_5} = \vec{a}_{F_6} + \vec{a}_{F_5F_6} + \vec{a}_{cor}$$

$$\rightarrow \vec{a}_E^n + \vec{a}_E^t + \vec{a}_{FE}^n + \vec{a}_{FE}^t = \vec{a}_{F_5F_6} + \vec{a}_{cor}$$

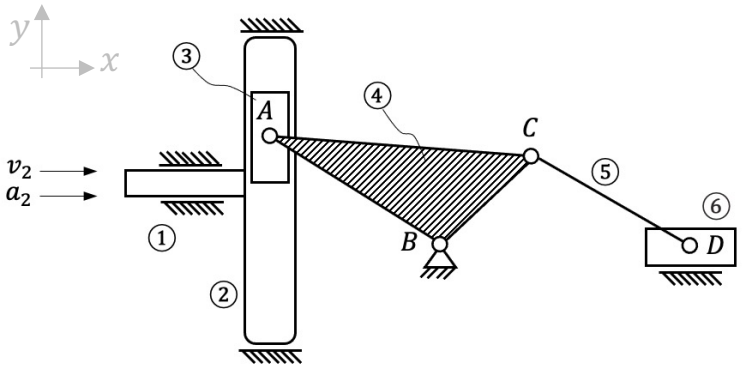
$$\omega_4^2\,\overrightarrow{EC} + \vec{\alpha}_4 \times \overrightarrow{CE} + \omega_5^2\,\overrightarrow{FE} + \vec{\alpha}_5 \times \overrightarrow{EF} = \vec{a}_{F_5F_6} + 2\vec{\omega}_5 \times \vec{v}_{F_5F_6}$$

Essendo $\vec{a}_{FE}^t = \vec{\alpha}_5 \times \overrightarrow{EF}$ $\rightarrow$ $\alpha_5 = \dfrac{|\vec{a}_{FE}^t|}{\overline{EF}}$

$$\alpha_6 = \alpha_5$$



MDP: scotch-yoke + manovellismo

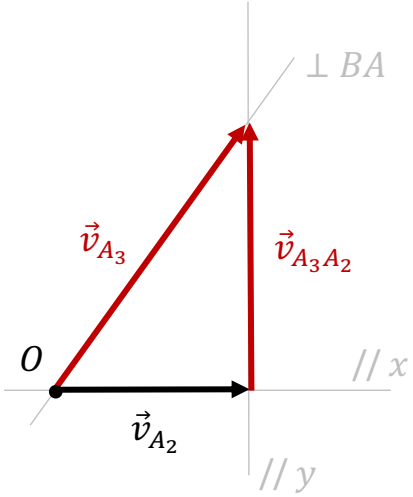


	MOD	DIR		MOD	DIR
v_{A_3}	?	$\perp \overline{BA}$	v_D	?	$// x$
v_{A_2}	v_2	$// x$	v_C	$\omega_4 BC$	$\perp \overline{BC}$
$v_{A_3A_2}$	?	$// y$	v_{DC}	?	$\perp \overline{DC}$

$$\vec{v}_{A_3} = \vec{v}_{A_2} + \vec{v}_{A_3A_2}$$

$$\vec{v}_{A_3} = \vec{v}_{A_4} = \vec{\omega}_4 \times \overrightarrow{BA}$$

$$\vec{\omega}_4 \times \overrightarrow{BA} = \vec{v}_{A_2} + \vec{v}_{A_3A_2}$$

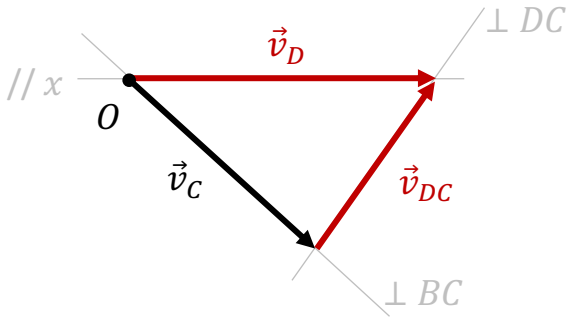


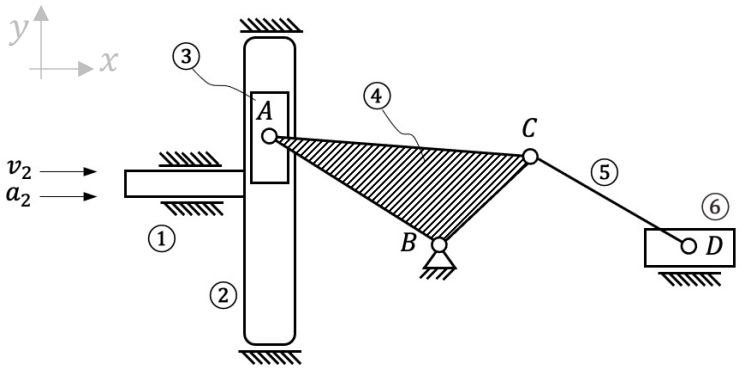
Essendo $\vec{v}_{A_3} = \vec{\omega}_4 \times \overrightarrow{BA}$ → $\omega_4 = \dfrac{|\vec{v}_{A_3}|}{\overline{BA}}$

$$\omega_2 = \omega_3 = 0$$

$$\vec{v}_D = \vec{v}_C + \vec{v}_{DC} \rightarrow \vec{v}_D = \vec{\omega}_4 \times \overrightarrow{BC} + \vec{\omega}_5 \times \overrightarrow{CD}$$

Essendo $\vec{v}_{DC} = \vec{\omega}_5 \times \overrightarrow{CD}$ → $\omega_5 = \dfrac{|\vec{v}_{DC}|}{\overline{CD}}$





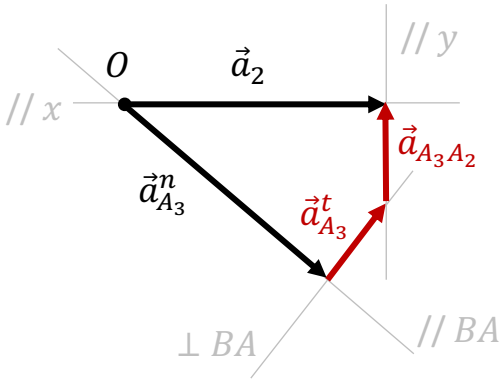
	MOD	DIR
$\vec{a}^{\,n}_{A_3}$	$\omega_4^2\ BA$	$//\overline{BA}$
$\vec{a}^{\,t}_{A_3}$	?	$\perp \overline{BA}$
$\vec{a}_{A_2}$	a_2	$// x$
$\vec{a}_{A_3A_2}$	?	$// y$

	MOD	DIR
$\vec{a}_D$	?	$// x$
$\vec{a}^{\,n}_C$	$\omega_4^2\ CB$	$//\overline{CB}$
$\vec{a}^{\,t}_C$	$\alpha_4\ CB$	$\perp \overline{CB}$
$\vec{a}^{\,n}_{DC}$	$\omega_5^2\ DC$	$//\overline{DC}$
$\vec{a}^{\,t}_{DC}$	?	$\perp \overline{DC}$

$$\vec{a}_{A_3} = \vec{a}_{A_2} + \vec{a}_{A_3A_2} + \vec{a}_{cor} \quad \rightarrow \quad \vec{a}^{\,n}_{A_3} + \vec{a}^{\,t}_{A_3} = \vec{a}_{A_2} + \vec{a}_{A_3A_2} + \vec{a}_{cor}$$

$$\omega_4^2\ \overrightarrow{AB} + \vec{\alpha}_4 \times \overrightarrow{BA} = \vec{a}_{A_2} + \vec{a}_{A_3A_2} + 2\vec{\omega}_3 \times \vec{v}_{A_3A_2}$$

Essendo $\vec{a}^{\,t}_{A_3} = \vec{\alpha}_4 \times \overrightarrow{BA}$ $\rightarrow$ $\alpha_4 = \dfrac{|\vec{a}^{\,t}_{A_3}|}{\overline{BA}}$



$$\vec{a}_D = \vec{a}_C + \vec{a}_{DC} \quad \rightarrow \quad \vec{a}_D = \vec{a}^{\,n}_C + \vec{a}^{\,t}_C + \vec{a}^{\,n}_{DC} + \vec{a}^{\,t}_{DC}$$

$$\vec{a}_D = \omega_4^2\ \overrightarrow{CB} + \vec{\alpha}_4 \times \overrightarrow{BC} + \omega_5^2\ \overrightarrow{DC} + \vec{\alpha}_5 \times \overrightarrow{CD}$$

Essendo $\vec{a}^{\,t}_{DC} = \vec{\alpha}_5 \times \overrightarrow{CD}$ $\rightarrow$ $\alpha_5 = \dfrac{|\vec{a}^{\,t}_{DC}|}{\overline{CD}}$

